

hardware float

we used Arduino nano as its compact size to achieve the most suitable design for the float. And we use pressure sensor bmp280 to know the deep of the float. And The input power is supplied by alkaline 9V batteries, to power the Arduino and regulator AMS1117 3.3V to power the and pressure sensor, NRF module and another one of 5V to power the Arduino nano and we used driver L298N and it will be able to provide the need power the dc motor used.

Float Power Calculations

Component	Input Voltage	Max Current	Quantity	Consumed Power
DC Motor	9v	500mA	1	4500mW
NRF	3.3v	12.3mA	1	40.59mW
nano	9v	40mA	1	360mW
Driver L298N	5v	36mA	1	180mW
Pressure sensor	3.3v	1.120 mA	1	3.696mW

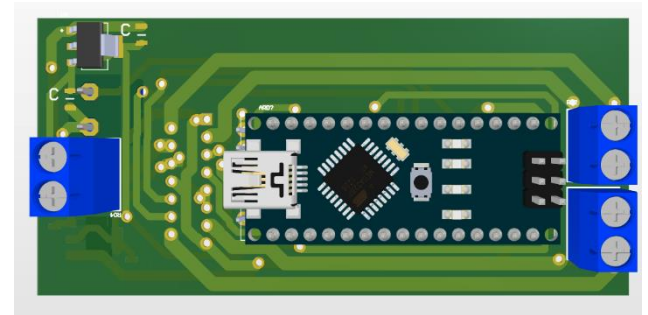


Figure 1 top view

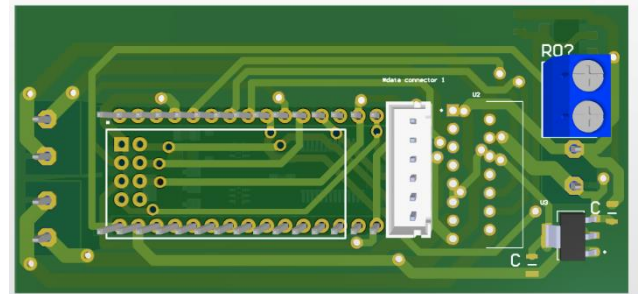


Figure 2 bottom view